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→ Computer network by Andrew S Tanenbaum.

→ Computernetwork :-

Two or more than two computer connect Each other and share Information and reshorce is known as Computer network.

→ Types of network :-
- LAN (Local area network)
- MAN (Metropolitan area network)
- WAN (wild area network)
- SAN (Storage area network)
- CAN (Campus area network)
- HAN (Home area network)
- PAN. (Personal are network).

→ needs of networking :-
- security
- Data sharing
- Resource sharing
- Communication efficiency
- Cost efficiency
- Scalability / Device add

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* History of Computer network.

1950 - 60's
ARPA (Advanced Research Project Agency)

1969
ARPA NET

1980
TCP / IP

1980's
ISO by OSI Model
OSI (Open System Interconnect)
ISO (International Standard of organization)

1990's
Internet → Inter - networking.

2000's
Mobile Network / wireless network.

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→ networking model :-
- OSI model
- TCP/IP model.

→ **OSI Model Layer** :-

1 - Physical layer → Physically connecting wifi / 5G / Ethernet etc.
2 - Data link layer → ensure node to node Commu
3 - network layer → manages routes & IP Add
4 - Transport TCP/UDP layer → Ensure end to end Commu
5 - session layer → manages Connection
6 - Presentation layer → O,I Encrypt compression
7 - Application layer → networking servies.

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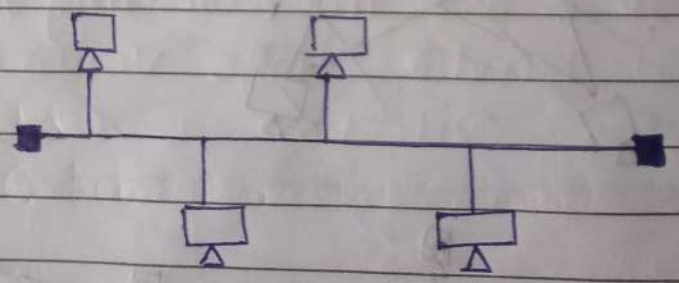
Types of Topology :-
- Bus
- Ring
- Tree
- Star
- Mesh
- Hybrid.

* what is Topology :-
- It is nothing but some kind of arrangement or we connect node from each other via network.

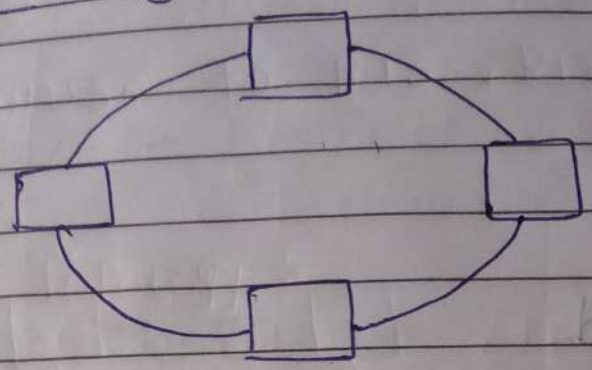
① Bus :-
In Bus Topology we can connect node via a single cable and it will easily manageable.



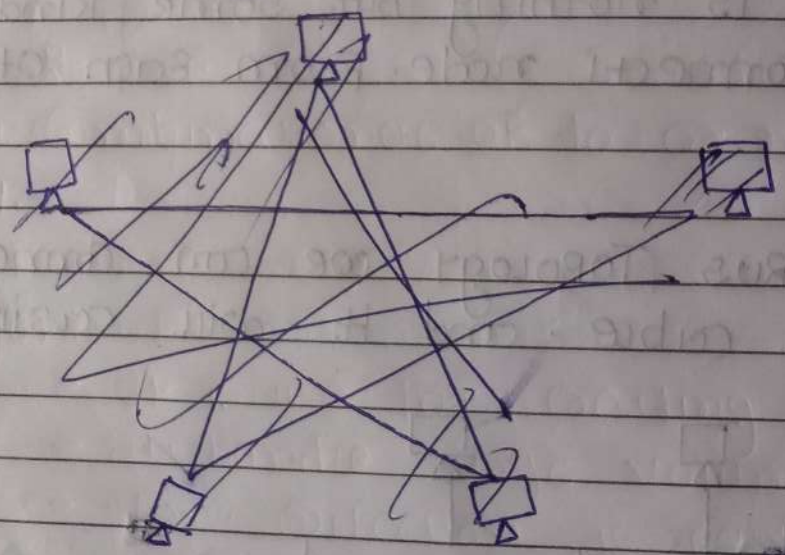
Advantage :- - easy to manage
- easily adding or deleting node.

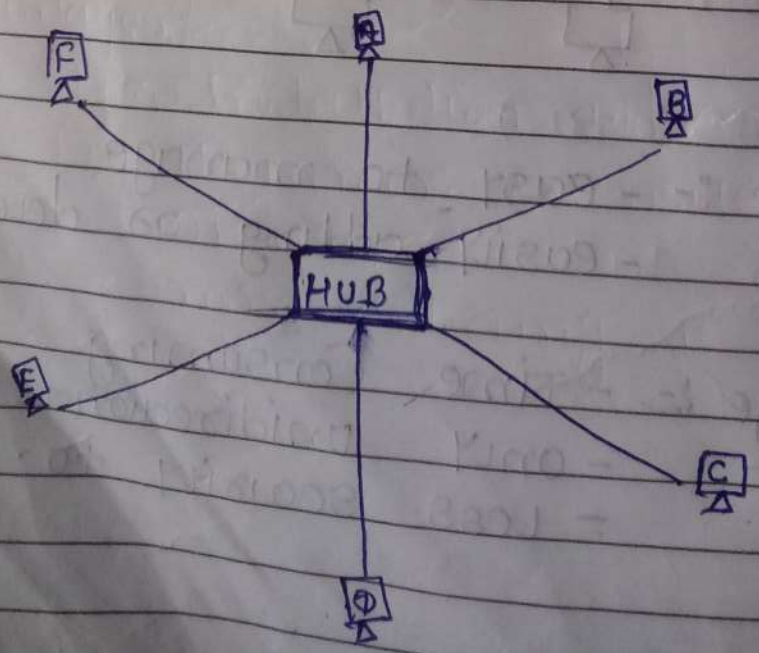
Disadvantage :- - Time Consuming
- only unidirectional
- Less security to other

② Ring Topology :-



③ Star Topology :-





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→ Advantages :-

- Adding as Removing nodes become easier as comes to Bus and Ring topology.

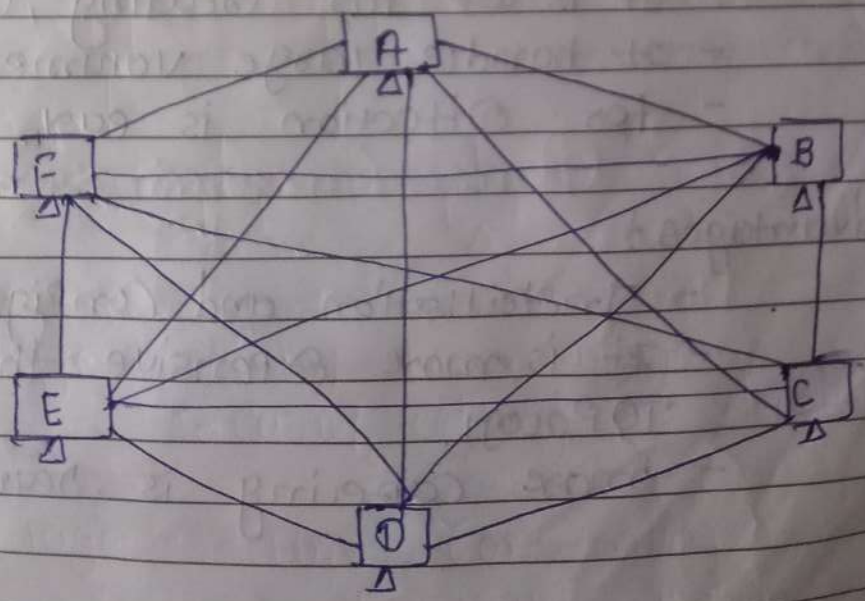
- If single node fails it will not lead to network.

- Troubleshoting becomes easier.

→ Disadvantages :-

- If Control.

④ Mesh Topology :-



star
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varify

→ Advantages :-

- Faster than Bus, Ring and Star Topology
- If single mode fail it will not lead to network failes.

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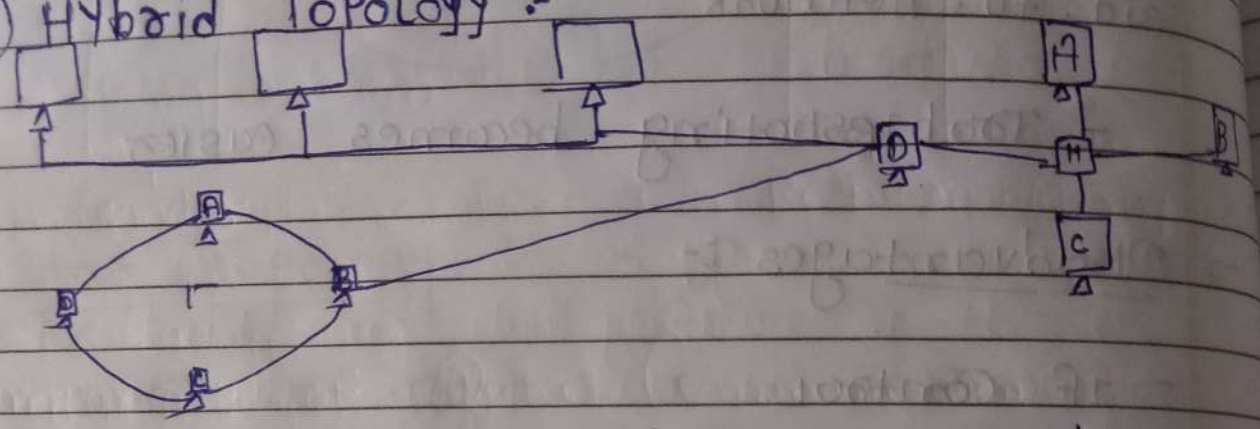
Disadvantages:-

- costly compure to BUS, Ring and Star.
- Complex

(5) Hybrid Topology :-



It is a Combinational of to or more different topology.

advantages :

- It is use for creating, Larger network
- It handle large volume of traffic
- For Direction is easy.

Disadvantages :

- Installation and Configuration Difficult
- It is more expensive then other TOPology.
- more cabeling is required.

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* network Architecture :-
  1. centralized
  2. Decentralized
  3. Peer to Peer
  4. client and server Model.

① centralized :-
All network resources and data are stored and manage by central server.

advantage :-
- easier manage control
- centralized services.

Disadvantage :- single point of failer.

② Peer to Peer :-
Each device both client and server Resources share directly between devices.

Ex :- Big torent networks.

Advantage :- Cost effective.
High scalability.

Disadvantage :
Difficulty Manage.

③ client and server model :-

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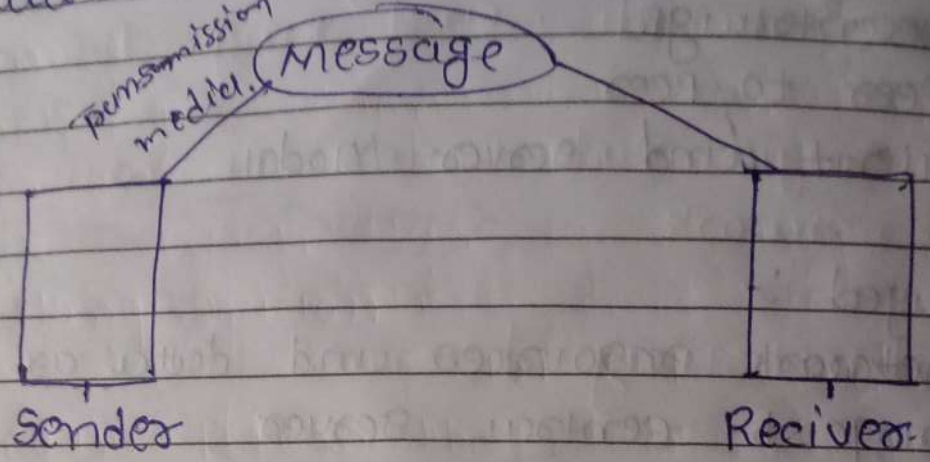
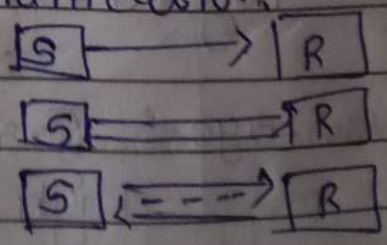
⇒ Data Communication :-



* Component of Data Communication
  - Message
  - Sender
  - Reciver
  - Transmission medium
  - Set of Rules [Protocol]

* Types of Data Communication
  - Simplex
  - Half Duplex
  - Full Duplex.



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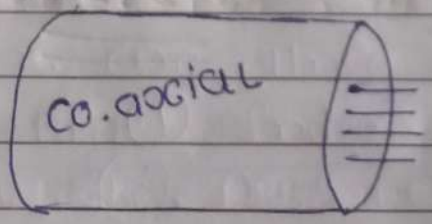
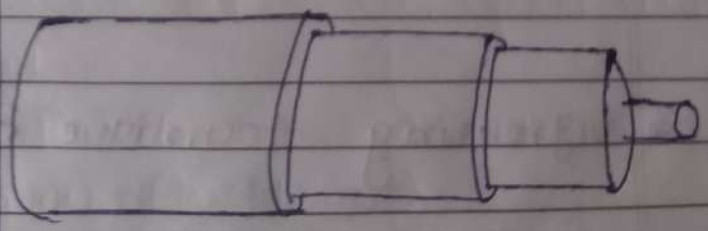
Transmission media

- Guided
  - Co-axial
  - Twisted Pair
    - ~~UTP~~ UTP (Unshelded)
    - STP (shelded).
  - Fiber optic
- unguided
  - Radio
  - micro-wave
  - Infrared.

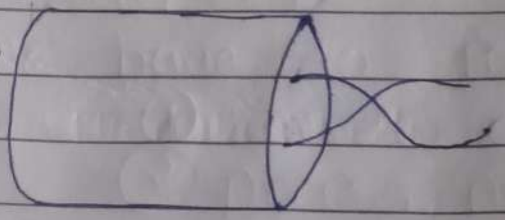
*

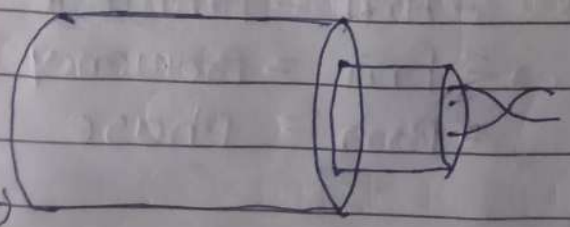


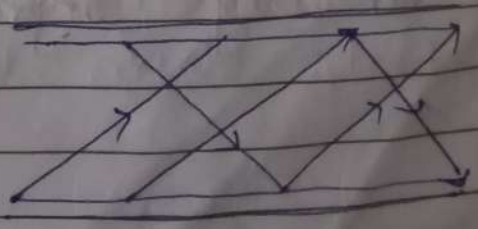


Fiber optic cable

Twisted Pair







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→ Data transfering technique :-

1. circuit switching
2. message switching
3. Packet switching.

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